23. Working with Instruments and Effects

Every track in Live can host a number of devices. These devices can be of three different sorts:

- *MIDI effects* act upon MIDI signals and can only be placed in MIDI tracks.
- *Audio effects* act upon audio signals and can be placed in audio tracks, return tracks, and the Main track. They can also be added to MIDI tracks as long as they are placed after an instrument.
- *Instruments* receive MIDI and output audio; they can be placed in MIDI tracks.

To learn about a particular device and how to operate it, consult the Live Audio Effect Reference, Live MIDI Effect Reference, or Live Instrument Reference chapters.

To learn about creating and using custom groupings of instruments and effects, check out the Instrument, Drum and Effect Racks chapter.

Further information about track types in Live can be found in the Routing and I/O chapter, including information on using return tracks to distribute the effect of a single device amongst several tracks.

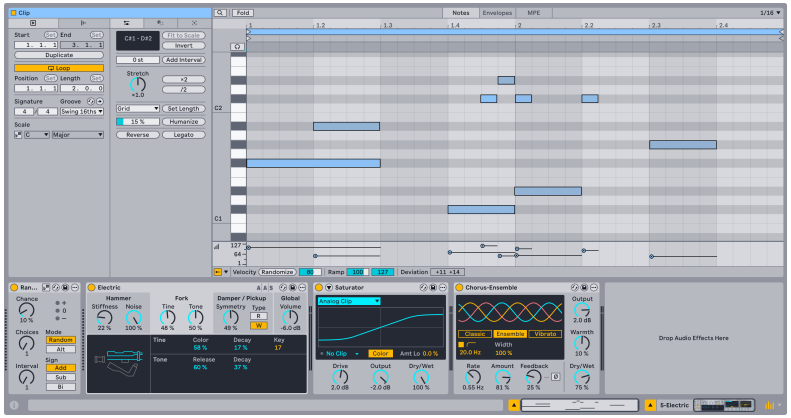
You can get hands-on with devices by assigning their parameters to MIDI or key remote control.

23.1 Device View

The Device View is where you insert, view, and adjust the devices for a selected track. You can double-click a track's title bar to view its devices in the Device View, which appears at the bottom of Live's window.

The Device View.

By default, either the Device View or the Clip View can be used at a time depending on whether a device or clip is in focus. If you want to tweak device parameters and edit notes or samples without switching between views, you can stack the Clip View above the Device View.



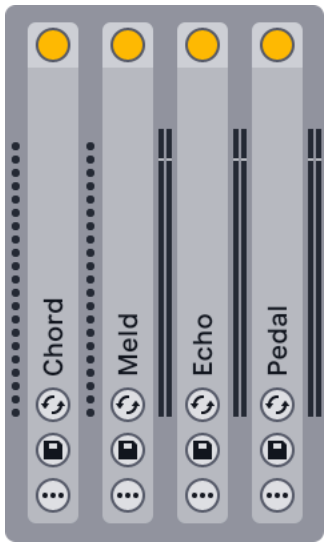
Stacked Clip and Device View.

Use the toggles next to the Clip and Device View Selectors in the bottom-right corner of Live's window to expand both views.

The Clip and Device View Selectors.

You can also use the keyboard shortcuts `Ctrl` `Alt` `3` (Win) / `Cmd` `Option` `3` (Mac) to show or hide the Clip View and `Ctrl` `Alt` `4` (Win) / `Cmd` `Option` `4` (Mac) to show or hide the Device View.

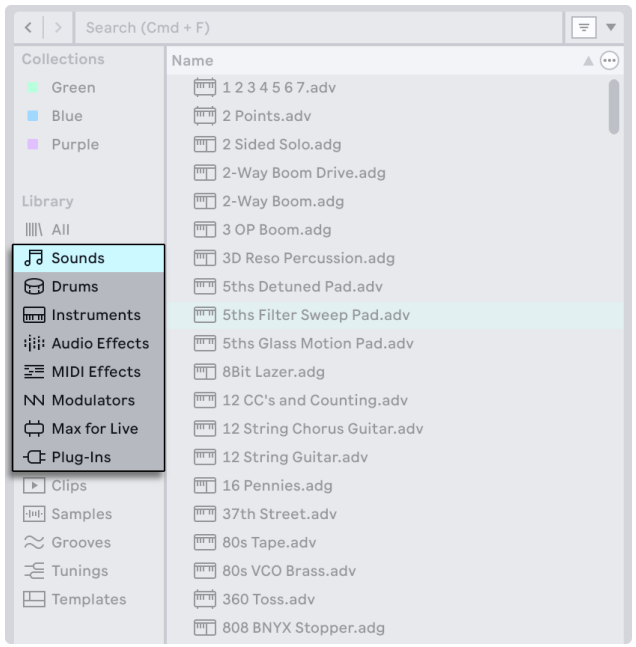
To save space in the Device View, a device can be collapsed by double-clicking its title bar or by choosing Fold from the title bar's context menu.



Devices Can Be Folded.

23.2 Using Devices

Live's devices can be accessed in the browser. The browser's sidebar contains labels for built-in devices, Max for Live devices, and third-party plug-ins.

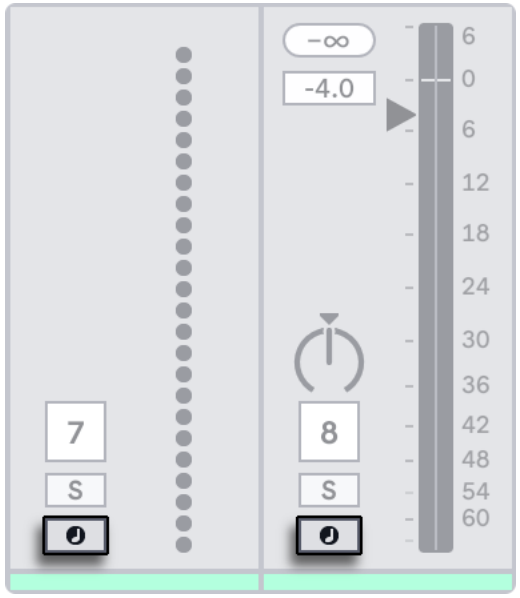


Device Labels in the Browser Sidebar.

The easiest way to place a device in a track is to double-click it in the browser, which adds it to the selected track or creates a new track if one is not selected. Alternatively, select a destination track by clicking within it, then select a device or preset in the browser and press `Enter` to add it to the selected track.

You can also drag devices into tracks or drop areas in the Session and Arrangement Views, or into the Device View. Dragging a sample to the Device View of a MIDI track creates a Simpler instrument with this sample loaded.

If you want to use an external input signal to feed a track in Live, the track's *Arm* button in the mixer must be activated in order to hear the input through the devices in the track's device chain when using the default Auto monitoring setting. On MIDI tracks, this is normally activated automatically when inserting an instrument.



MIDI and Audio Track Arm Buttons.

This is how you would play live instruments through effects on a track, for example, or use a MIDI keyboard's input to play a track's instrument. You can easily move from this setup into recording new clips for further use in a Set. If you have alternative monitoring preferences, please see the Monitoring section to learn how to configure these settings.

To add another device to the track, simply drag it there or double-click its name to append it to the device chain. Signals in a device chain always travel from left to right.

You can drop audio effects in at any point in an audio track's device chain, keeping in mind that the order of effects determines the resulting sound. The same is true for a MIDI track's device chain.

If you drop an instrument into a MIDI track's device chain, be aware that signals following (to the right of) the instrument are *audio signals*, available only to audio effects. Signals preceding (to the left of) the instrument are *MIDI signals*, available only to MIDI effects. This means that it's possible for a MIDI track's device chain to hold all three types of devices: first MIDI effects, then an instrument, and finally audio effects.

A MIDI Track's Device Chain Can Contain All Three Device Types.

To remove a device from the chain, click on its title bar and press your computer's `Backspace` or `Delete` key, or select Delete from the Edit menu. To change the order of devices, drag a device by its title bar and drop it next to any of the other devices in the Device View. Devices can be moved to other tracks entirely by dragging them from the Device View into the Session or Arrangement Views.

Edit menu commands such as cut, copy, paste, and duplicate can be used on devices. Pasted devices are inserted in front of the selected device. You can paste at the end of a device chain by clicking in the space after the last device, or by using the right arrow key to move the selection there. Generally, devices can be placed, reordered, and deleted without interrupting the audio stream.

Devices in Live's tracks have input and output level meters. These meters are helpful in finding problematic devices in the device chain: low or absent signals will be revealed by the level meters, and relevant device settings can then be adjusted, or the device can be turned off or removed.

The Level Meters Between Devices in a Chain.

Note that no clipping can occur between devices because there is practically unlimited headroom. Clipping can occur when an overly strong signal is sent to a *physical output* or written to a sample file.

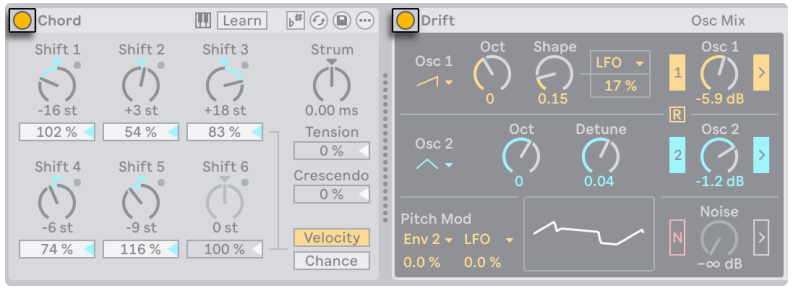
After reading about using devices in Live, you might find it interesting to look into clip envelopes, which can automate or modulate individual device parameters on a per-clip basis.

23.2.1 Device Title Bar

The controls in a device's title bar can be used to activate or deactivate the device, save and hot-swap presets, create default presets, and open the device's context menu.

Certain title bar controls are only available for some devices, such as the scale awareness toggle for devices that can follow a clip's scale, and the Learn toggle for the Chord device.

Devices are turned on or off using their Activator toggles.



Device Activator Toggles.

Turning a device off is like temporarily deleting it: the signal remains unprocessed, and the device does not consume CPU cycles. Live devices generally have no impact on CPU usage unless they are active. For more information, please refer to the Managing the CPU Load section. The Freeze Track command discussed there is especially helpful when working with CPU-intensive devices.

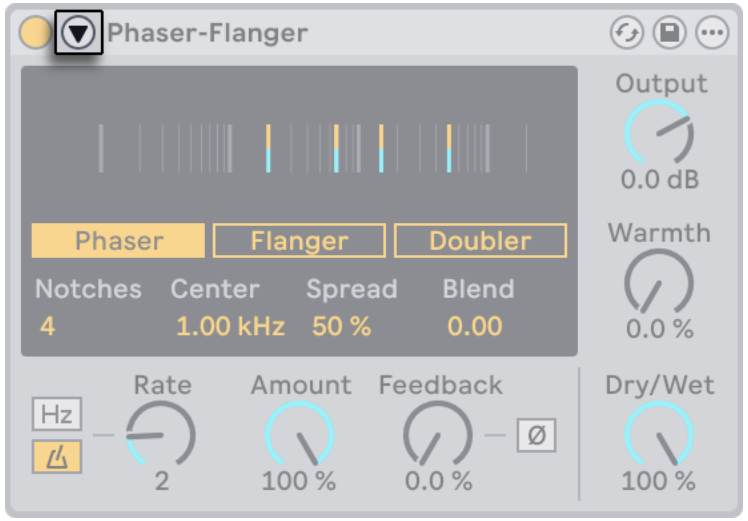
Some devices have views that you can expand above the Device View, such as the Gain Stage and Modulation Matrix sections in Roar and the Frequency Display in EQ Eight.

Devices with this kind of view have an arrow toggle to the right of the Activator toggle. Use the toggle to show or hide the view.

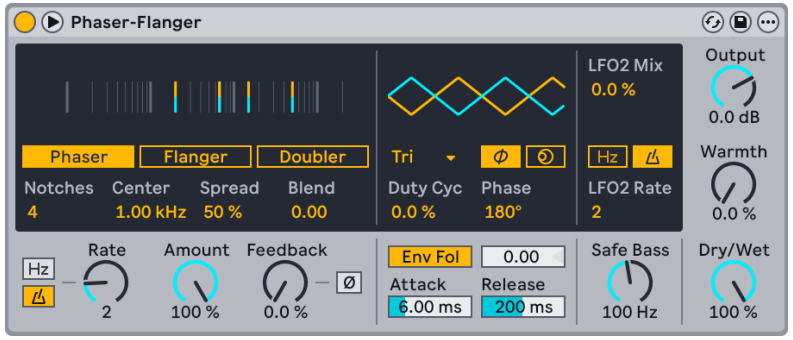
The Expanded View Toggle for Roar.

Roar's Expanded View.

Some views can also be expanded within a device itself, for example, the LFO and Envelope Follower sections in Phaser-Flanger. This kind of view is indicated by a triangle icon in the device title bar.



The Expanded View Toggle for Phaser-Flanger.

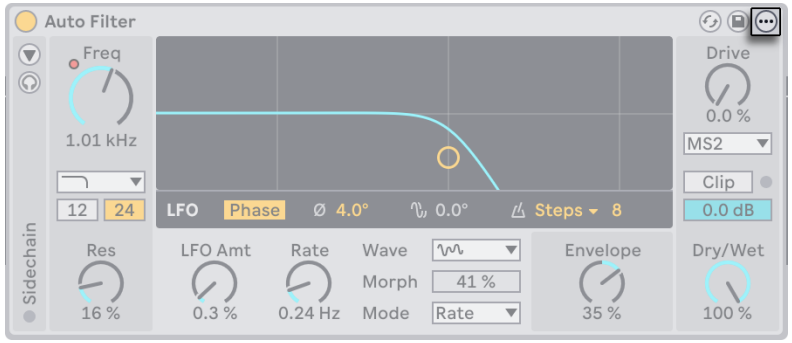


Phaser-Flanger's Expanded View.

A context menu is available for every device and contains general editing commands such as Cut, Copy, and Rename. Some commands are unique to a particular device, such as the Mono Sidechain option in Auto Filter.

Context Menu Options for Auto Filter.

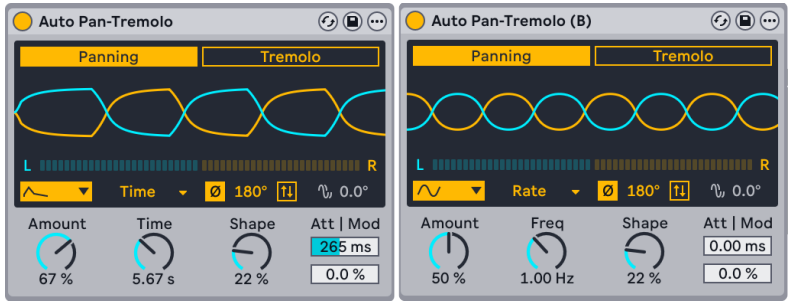
You can right-click a device's title bar or use the Show Options toggle to open the context menu.



The Show Options Toggle.

23.2.2 Device A/B Comparison

Every built-in Live device includes two device states, A and B, which can store separate parameter values. This lets you save and compare the changes you make when creating or editing presets. You can save an iteration of a preset that you don't want to overwrite to one device state and continue experimenting on the other.



A Device Can Store Two Sets of Parameter Values.

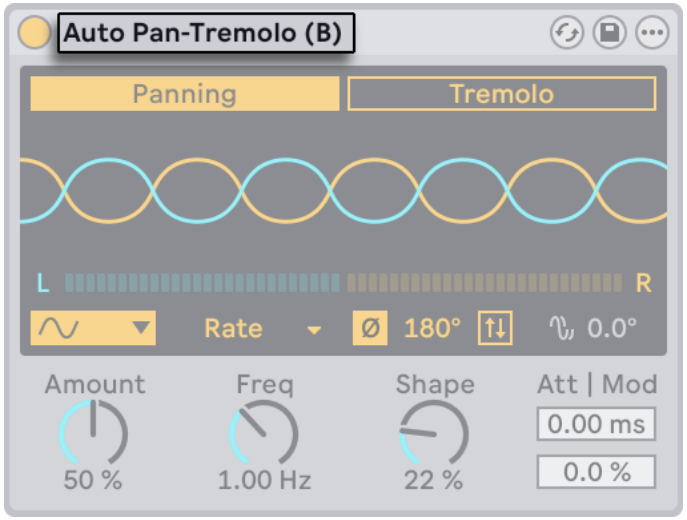
Using device states to compare parameter settings can also be useful for things like EQing and compression, where minor adjustments can make a noticeable difference. You can try out slight parameter variations to hear how they affect the resulting sound or check the changes against default values.

When a device is first loaded, its default parameter values are used for both A and B. Once you adjust a parameter, B retains the initial values, and any changes you make only apply to A.

To copy the current parameter values from the A device state to B, select the *Compare: Copy A to B* command from the Edit menu, the device's context menu, or its Show Options toggle.

You Can Copy the Parameter Values from One Device State to Another.

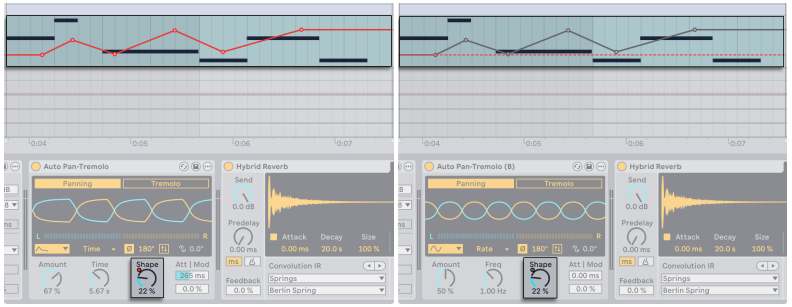
You can switch from the A device state to B using the *Compare: Switch to B* command or the P key. When the B device state is selected, a "(B)" appears next to the device's name in the title bar.



The B Device State.

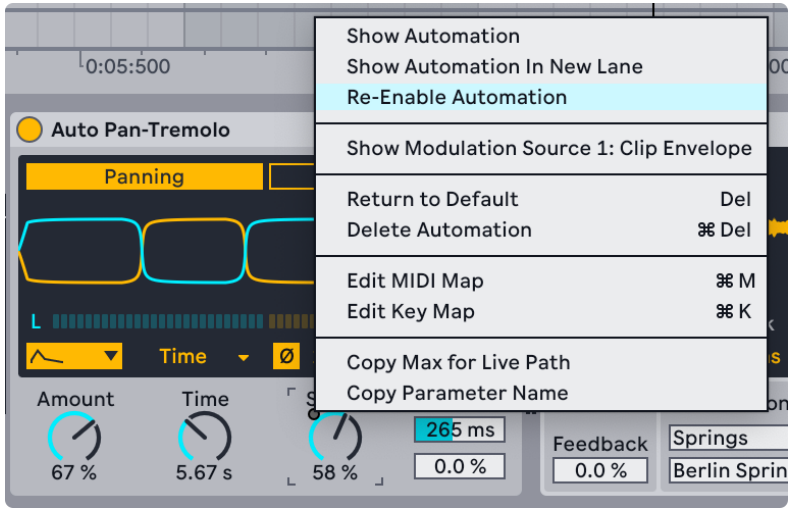
The Compare commands are available for both device states, so how you save parameter values to A and B is up to you. You could use one device state to keep the default values as a reference, or use both states to design completely different-sounding presets. Note that these commands are not available for Racks, Max for Live devices, or plug-ins.

Because A and B are unique device states, any parameter automation you create in one state is not shared with the other. This means if you automate a parameter in the A device state, that automation will be disabled when you switch to B and is not automatically re-enabled when you switch back to A.



Automation Is Disabled When Switching Between Device States.

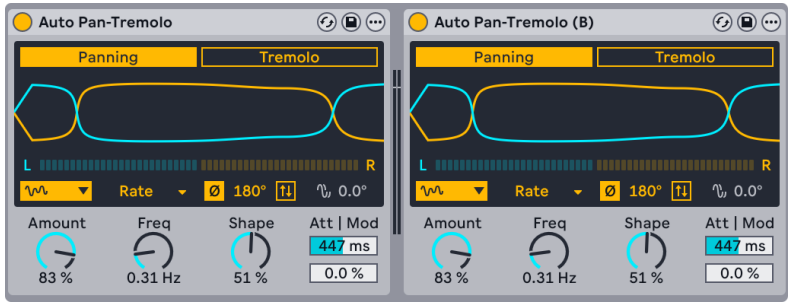
To reactivate any disabled parameter automation, use the *Re-Enable Automation* command in the parameter's context menu when the device state you want to use is selected.



Re-Enable Automation Using the Context Menu Command.

This command can be used from either device state. Keep in mind that since automation is specific to A and B, it will be disabled again each time you switch from one state to the other.

Only the parameter values from the selected device state are copied or duplicated. For example, if you copy a device while A is selected, the new device will use the parameter values from the original's A device state for both A and B, even if the original's B device state has other values.



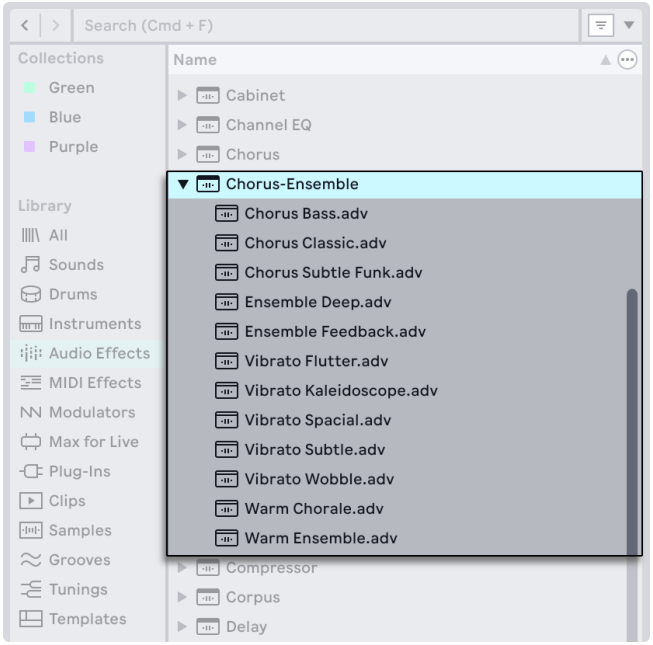
Only the Parameter Values from a Selected Device State (Left) Are Copied (Right).

Similarly, if you save a preset to your User Library, the parameter values from the selected device state are used for both A and B.

Note that device states can be accessed via the Live API with Max for Live.

23.2.3 Live Device Presets

Live devices can store and retrieve their parameter settings as *presets*. Each device appears in the content pane of the browser as a folder that can be opened to reveal its presets.



Presets in the Browser.

The folder represents the default preset for a device; this can be either the original factory settings or default settings you have saved. The presets within the folder have custom settings. Presets are included as part of the Core Library and Packs. You can also create and save your own presets.

You can browse and load presets quickly with the computer keyboard:

- Scroll up and down using the up and down arrow keys.
- Close and open device folders using the left and right arrow keys.
- Press `Enter` to load a preset onto a selected track.

You can also double-click a preset in the browser to load it onto a selected track, or drag and drop a preset from the browser onto a track's title bar or device chain.

Note that it is generally recommended to load presets via the browser; however, it is also possible to drag Live-specific device file types (such as .adg, .adv, and .amxd) directly into a Live Set from the Explorer (Win) / Finder (Mac).

If you want to replace a preset on a track with a different one, you can drag and drop a new preset from the browser and place it over the existing preset in the track's device chain.

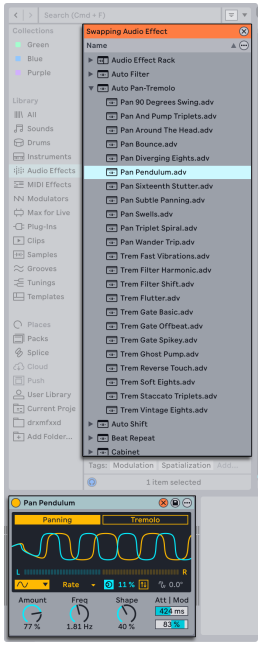
23.2.4 Hot-Swapping Presets

Using *Hot-Swap mode* is a quick way to replace a preset on a track. In this mode, the preset on the track is linked to the browser so that you can audition and load different presets. To start hot-swapping, press `Q` or click the *Hot-Swap Presets* button in the device's title bar.

The Hot-Swap Presets Button.

If you press `Q` to enter Hot-Swap mode, the currently selected preset on the track will be made available for swapping. If no preset is selected, the first audio effect (on audio tracks) or the instrument (on MIDI tracks) will be swapped.

Once Hot-Swap mode is enabled, the preset is selected in the browser and its device folder is expanded. You can replace the preset with another one from the same folder or navigate to a different folder to swap in a preset for another device. Only presets of the same device type can be hot-swapped. For example, it is not possible to swap an audio effect for a MIDI effect.



Browse and Replace Presets in Hot-Swap Mode.

Use the up and down arrow keys to navigate through the presets. Press `Enter` to load the selected preset or double-click the preset you want to load. If you want to use a device's default settings, select the device's parent folder and load that instead of one of its presets.

Note that you can automatically load presets in Hot-Swap mode by adding the -EnableHotSwapOnSelection entry to the Options.txt file. With this option, the existing preset is replaced as soon as a new one is selected in the browser.

Once you are done replacing presets, you can exit Hot-Swap mode with any of the following actions:

- Press the `Q` key.
- Press the `Esc` key.
- Click the X button in the Hot-Swap bar at the top of the browser's content pane.
- Click the X button in the device's title bar.
- Navigate from the browser to a different view.

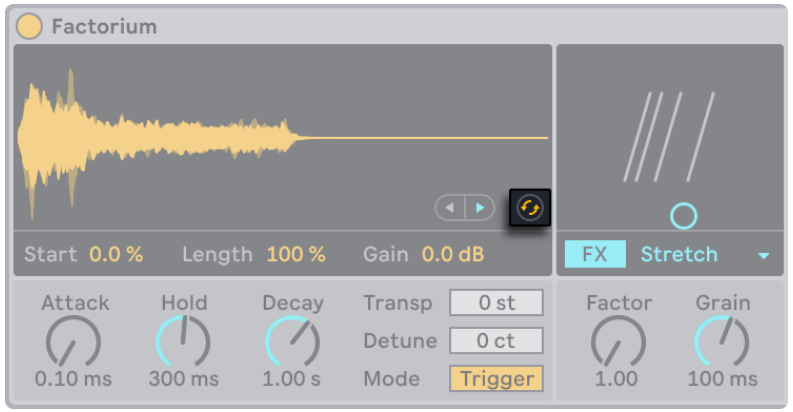
23.2.4.1 Hot-Swapping Samples

In addition to replacing entire presets, you can swap out loaded samples in Drum Rack pads, Drum Sampler, Impulse, Sampler, and Simpler.

To do so, use the *Hot-Swap Sample* button that appears in the following locations:

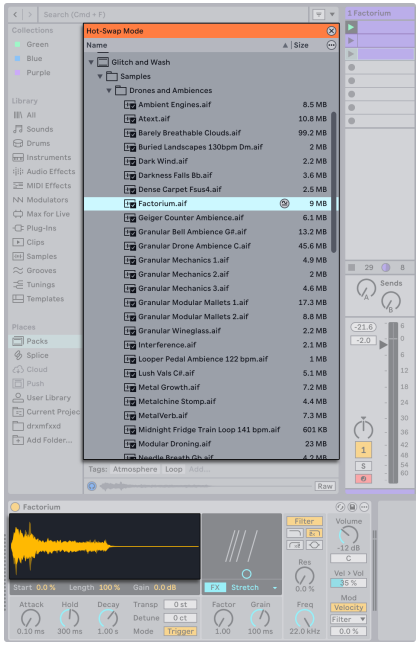
- Drum Rack pads — At the bottom right of each pad.
- Drum Sampler — At the bottom right of the Waveform Display.
- Impulse — At the bottom right of each sample slot.
- Sampler — At the top left of the Zone Editor when it is expanded.
- Simpler — At the bottom right of the Sample Display.

Note that the button is only visible when you hover over the relevant area (except in Sampler, where it is always visible). For example, you can hover over the Waveform Display to access the button in Drum Sampler.



The Hot-Swap Sample Button for Drum Sampler.

Clicking the button opens the folder where the sample is located in the browser. From there, you can replace the loaded sample with another one from the same folder or swap it with a sample from a different folder. Press `Enter` to load a selected sample or double-click the sample you want to load.



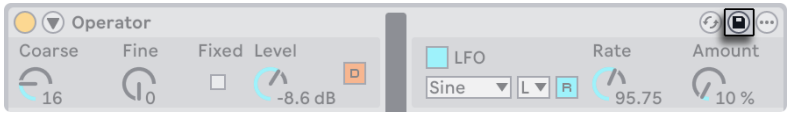
The Loaded Sample in a Device Can Be Swapped in the Browser.

When you're ready to stop swapping samples, you can exit Hot-Swap mode by:

- Clicking the Hot-Swap Sample button in the device.
- Clicking the X button in the Hot-Swap bar at the top of the browser's content pane.
- Pressing the `Esc` key.
- Pressing the `Q` key.
- Navigating from the browser to a different view.

23.2.5 Saving Presets

You can create and save any number of your own presets in the browser's *User Library*.



The Save Preset Button.

Click the Save Preset button to save a device's current settings (including any custom info text) as a new preset. You will be redirected to the browser, where you can press `Enter` to use Live's suggested name, or you can type one of your own. To cancel preset saving, press the `Esc` key. You

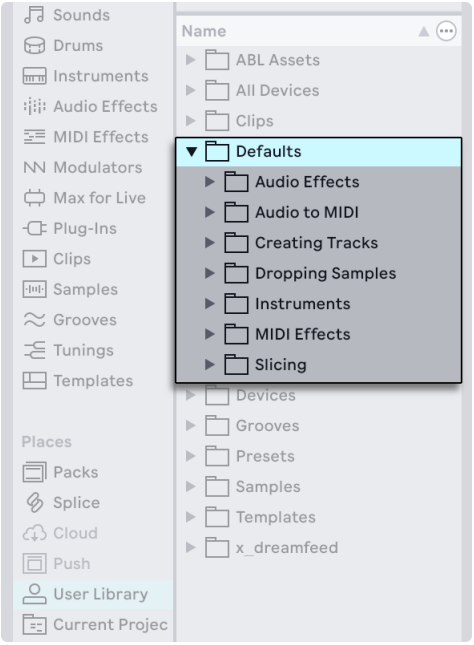
can also save presets to specific folders in the Places section of the browser (such as your Current Project folder) by dragging from the title bar of the device and dropping into the browser location of your choice.

For detailed information on what can be done with the browser, check out the Working with the Browser chapter. For more on how to store Project-specific presets, see the Projects and Presets section in the Managing Files and Sets chapter.

23.2.6 Default Presets

Presets saved to the *Defaults* folders in your User Library will load in place of the generic device settings. There are also Defaults folders that allow you to:

- Customize how Live responds to various user actions, such as sample dropping, slicing, and converting audio to MIDI.
- Cause newly created MIDI and audio tracks to load with certain devices already in place, complete with custom parameter settings.
- Load VST and Audio Units plug-ins with a specific collection of parameters already configured in Live's panel.



The Default Presets folders in the User Library.

To save the current settings of a Live device as a default preset, open the context menu on the device's header and select "Save as Default Preset." This works for all of Live's instruments, MIDI effects, and audio effects (including the various types of Racks). If you have already saved a default preset for a particular device, Live will ask you before overwriting it.

To create a default configuration preset for a VST or Audio Unit plug-in:

- Load the selected plug-in to a track.
- From the plug-in's Configure Mode, set up your desired collection of parameters.
- Open the context menu on the track header and select "Save as Default Configuration."

If you have both VST and Audio Units versions of a particular plug-in installed, you can create separate default configuration presets for each type. Note that default presets for plug-ins do not save the *settings* of the configured parameters; only the parameter configuration within Live's panel is saved.

To create default presets for MIDI and audio tracks:

- Load the device(s) you would like as default onto a track (or no devices, if you would like your default track to be empty).
- Adjust the device parameters as you like.
- Open the context menu on the track header and select "Save as Default Audio/MIDI Track."

To specify how Live behaves when dragging a sample to a Drum Rack or the Device View of a MIDI track:

- Create an empty Simpler or Sampler.
- Adjust the parameters as you like.
- Drag the edited device to the "On Drum Rack" or "On Device View" folder, which can be found in the Defaults/Dropping Samples folders in your User Library. Drum Rack pad defaults can also be saved via a context menu option on a Drum Rack's pad.

To adjust how Live behaves when slicing an audio file:

- Create an empty Drum Rack.
- Add an empty Simpler or Sampler to the Drum Rack to create a single chain.
- Add any additional MIDI or Audio Effects to this chain.
- Adjust parameters in any of the devices.
- Assign Macro Controls to any of the controls in the chain's devices.
- Drag the entire Drum Rack to the Defaults/Slicing folder in your User Library.

You can create multiple slicing presets and choose between them in the Slicing Preset chooser in the slicing dialog.

To create default presets for converting Drums, Harmony, or Melody to MIDI:

- Create a MIDI track containing the instrument you would like to use as your default for a particular conversion type. Note that default presets for converting drums must contain a Drum Rack.
- Add any additional MIDI or Audio Effects to the track.

- Adjust parameters in any of the devices.
- If you're using multiple devices, group them into a Rack.
- Drag the entire Rack to the appropriate folder in Defaults/Audio to MIDI in your User Library.

In addition to these program-wide default presets, you can also create default presets that are specific to only one Project. This can be useful if, for example, you're using specialized device or track configurations for a particular Set, and would like to create variations of the Set which will also have access to these presets, but without overwriting the more generalized default presets you use for your other types of work. To create Project-specific default presets:

- Recreate the Defaults folder and any desired subfolders within the Project folder.
- Depending on which types of Project-specific defaults you'd like to work with, adjust the corresponding device parameters, track settings, etc.
- Save the device or track to the appropriate folder in your Project-specific Defaults folder.

Now, whenever you've loaded a Set from this Project, any default presets that you've saved into these Project folders will be used instead of those found in the User Library. Note that the context menu options for saving default presets will save them to your main User Library, and so cannot be used to save Project-specific defaults.

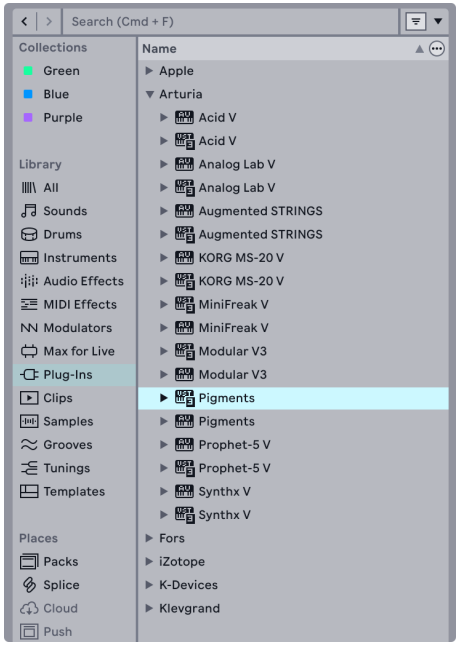
23.3 Using Plug-Ins

The collection of devices that you can use in Live can be extended with *plug-ins*. Live supports Steinberg Media's VST plug-in format, as well as the Audio Units (AU) plug-in format (macOS only), specifically:

- VST2
- VST3
- Audio Units 2
- Audio Units 3 (Live 11.2 and later)

Working with VST and Audio Units plug-ins is very much like working with Live devices. VST and AU instruments can only be placed in Live MIDI tracks and, like Live instruments, they will receive MIDI and output audio signals. Plug-in audio effects can only be placed in audio tracks or following instruments. Please see the previous section, Using the Live Devices, for details.

To skip plug-in scanning when Live is launched, hold the `Alt` (Win) / `Option` (Mac) modifier when opening the program until the splash screen closes. This can be helpful when troubleshooting crashes to see if any plug-ins are causing the problem.



Plug-Ins in the Browser.

Audio Units and VST plug-ins are browsed and imported using the browser's Plug-Ins label. Plug-in instruments can be differentiated from plug-in effects in the browser, as they appear with a keyboard icon.

Note that plug-in presets are only available in the browser for Audio Units plug-ins. In some instances, factory presets for Audio Units will only appear in the browser once the device has been placed in a track and its Hot-Swap button activated.

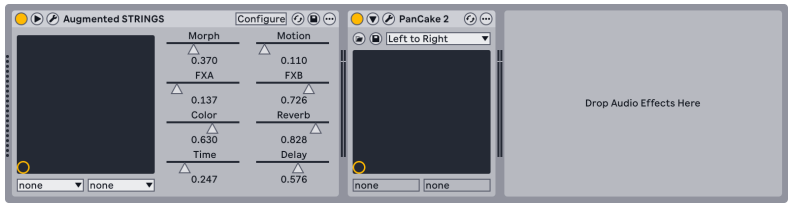
Note: The first time you start Live, no plug-ins will appear in the Plug-Ins label as you must first "activate" your plug-in sources. Activating your plug-in sources tells Live which plug-ins you want to use and where they are located on your computer. Information on activating (and deactivating) plug-in sources can be found later in this chapter, in the sections on the VST Plug-In folder and Audio Units Plug-Ins.

Note for "Intel® Mac" users: Intel® Mac computers cannot natively run VST or AU plug-ins that have been written for the PowerPC platform. Only plug-ins of type Universal or Intel® can be used in Live.

If you install/uninstall a plug-in while the program is running, Live will not detect your changes or implement them in the browser until the next time you start the program. Use the *Rescan* button in the Plug-Ins Settings to rescan your plug-ins while Live is running, so that newly installed devices become immediately available in the browser.

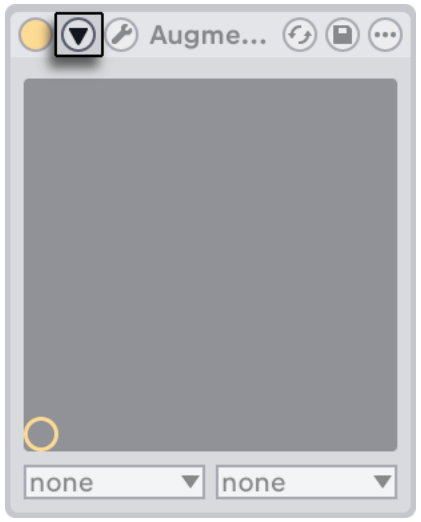
You can also rescan if you believe that your plug-in database has somehow become corrupted. Holding down the `Alt` (Win) / `Option` (Mac) modifier while pressing Rescan will delete your plug-in database altogether and run a clean scan of your plug-ins.

23.3.1 Plug-Ins in the Device View



VST Plug-Ins in the Device View.

Once a plug-in is dragged from the browser into a track, it will show up in the Device View. For plug-ins with up to 64 modifiable parameters, a Live panel will represent all of the parameters as horizontal sliders. Plug-ins that contain more than 64 parameters will open with an empty panel, which you can then configure to show the parameters you want to access. The plug-in's original interface can be opened in a separate window.

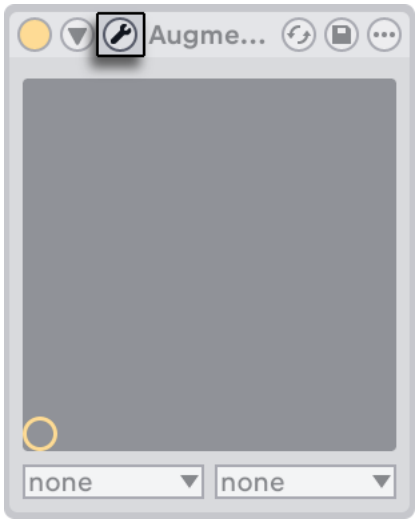


The Plug-In Unfold Button.

You can view or hide the plug-in's parameters by toggling the ▼ button in the plug-in's title bar.

The X-Y control field can be used to control two plug-in parameters at once and is therefore especially well-suited for live control. To assign any two plug-in parameters to the Live panel X-Y field, use the drop-down menus directly beneath it.

23.3.1.1 Showing Plug-In Panels in Separate Windows



The Show/Hide Plug-In Window Button.

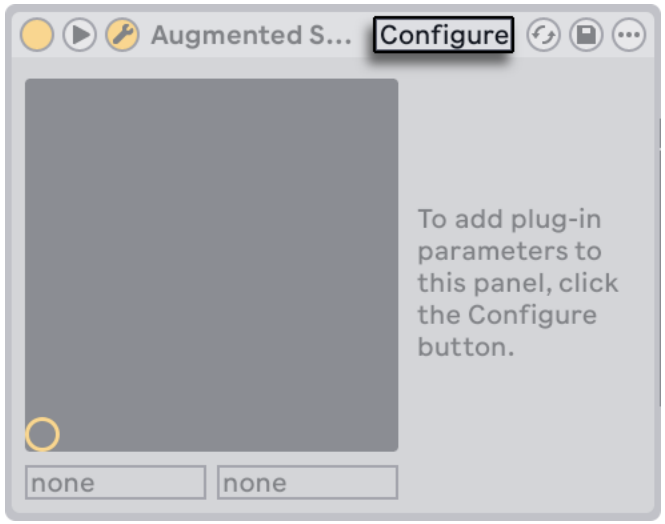
The Show/Hide Plug-In Window button opens a floating window that shows the original VST or Audio Units plug-in panel. Changing parameters on the floating window has the same effect as changing them in the Live panel, and vice versa.

There are a few important Plug-Ins Settings for working with plug-in windows:

- If activated, the *Auto-Open Plug-In Windows* option assures that plug-in windows open automatically when plug-ins are loaded into tracks from the browser.
- If the *Multiple Plug-In Windows* option in the Plug-In Settings is activated, you can open any number of plug-in windows at once. Even with this option off, you can hold down the `Ctrl` (Win) / `Cmd` (Mac) modifier when opening a new plug-in window to keep the previous window(s) from closing.
- Using the *Auto-Hide Plug-In Windows* option, you can choose to have Live display only those plug-in windows belonging to the track that is currently selected.

You can use the View menu's Show/Hide Plug-In Windows command or the `Ctrl` `Alt` `P` (Win) / `Cmd` `Option` `P` (Mac) shortcut to hide and show open plug-in windows. Note that the shortcut only works if the plug-in windows have already been opened manually first or if the Auto-Open Plug-In Windows option is enabled in Live's Settings. An open plug-in window shows the name of the track to which the plug-in belongs in the title bar of the plug-in window.

23.3.1.2 Plug-In Configure Mode



The Configure Button.

Configure Mode allows you to customize Live's panel to show only the plug-in parameters that you need to access. To do this:

- Enter Configure Mode by pressing the "Configure" button in the device's header.
- Click on a parameter in the plug-in window to add it to Live's panel. For some plug-ins, it may be necessary to actually change the parameter's value. Additionally, certain plug-ins do not "publish" all of their parameters to Live. These parameters cannot be added to Live's panel.

While in Configure Mode, parameters in Live's panel can be reordered or moved by dragging and dropping them to new locations. Parameters can be deleted by pressing the `Delete` key. If you try to delete a parameter that has existing automation data, clip envelopes, or MIDI, key or Macro mappings, Live will warn you before proceeding.

The parameters that you assign are unique for each instance of a given plug-in in your Set, and are saved with the Set. If you would like to save a setup using a particular collection of parameters, you can create a Rack containing the configured plug-in. Racks can then be saved to your User Library and loaded into other Sets. You can also save a particular parameter configuration as a default preset.

Certain plug-ins do not have their own windows, and instead only show their parameters in Live's panel. For these plug-ins, it is not possible to delete parameters when in Configure Mode (although they can still be moved and reordered).

There are several ways to add plug-in parameters to Live's panel without entering Configure Mode:

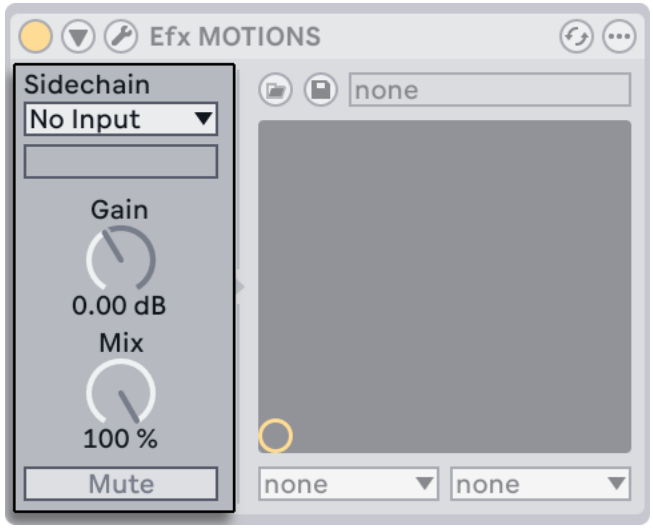
- Adjusting a parameter in the plug-in's floating window creates temporary entries for that parameter in the clip envelope and automation choosers, as well as the choosers in the panel's X-Y field. These entries are removed when you adjust another parameter. To make the entry permanent (thus adding it to Live's panel), either edit the parameter's automation or clip envelope, select another parameter in the automation or clip envelope choosers, or select the temporary parameter in one of the X-Y field's choosers.
- When a parameter is changed on a plug-in's floating window during recording, automation data is recorded automatically. When recording is stopped, the automated parameters are automatically added to Live's panels for any plug-ins that were adjusted.
- When in MIDI, key or Macro mapping mode, adjusting any parameter in the plug-in's window will create it in Live's panel. The new panel entry will be automatically selected, allowing you to map it immediately.

Once a plug-in is placed in a track and you have (optionally) configured its parameters in Live's panel, you can use it just like a Live device:

- You can map MIDI controller messages to all of the parameters in Live's panel.
- You can drag or copy the device to different locations in the device chain or to other tracks, according to the rules of audio effects and instruments.
- You can automate or modulate its continuous parameters with clip envelopes.
- You can use the multiple I/O features of some plug-ins by assigning them as sources or targets in the routing setup of tracks. See the Routing and I/O chapter for details.
- You can create custom info text for the plug-in.

23.3.2 Sidechain Parameters

Normally, the signal being processed and the input source that triggers the plug-in device are the same signal. But by using sidechaining, it is possible to apply processing to a signal based on the level of another signal. In plug-in devices that support sidechaining, you can access the sidechain parameters on the left side of the device.



Plug-In Sidechain Parameters.

The choosers allow you to select any of Live's internal routing points. This causes the selected source to act as the device's trigger, instead of the signal that is actually being processed.

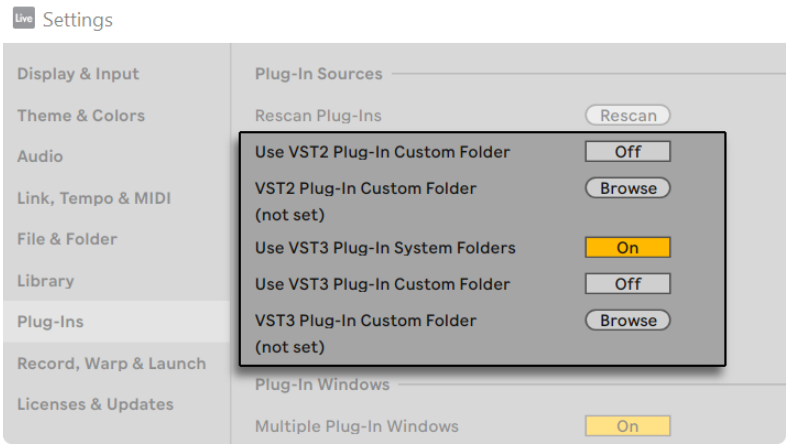
The Gain knob adjusts the level of the external sidechain's input, while the Mix knob allows you to use a combination of sidechain and original signal as the trigger. With Mix at 100%, the device is triggered entirely by the sidechain source. At 0%, the sidechain is effectively bypassed. Note that increasing the gain does not increase the volume of the source signal in the mix. The sidechain audio is only a trigger for the device and is never actually heard.

The Mute button allows you to listen to only the plug-in device's output, bypassing the sidechain source's input.

23.4 VST Plug-Ins

23.4.1 The VST Plug-In Folder

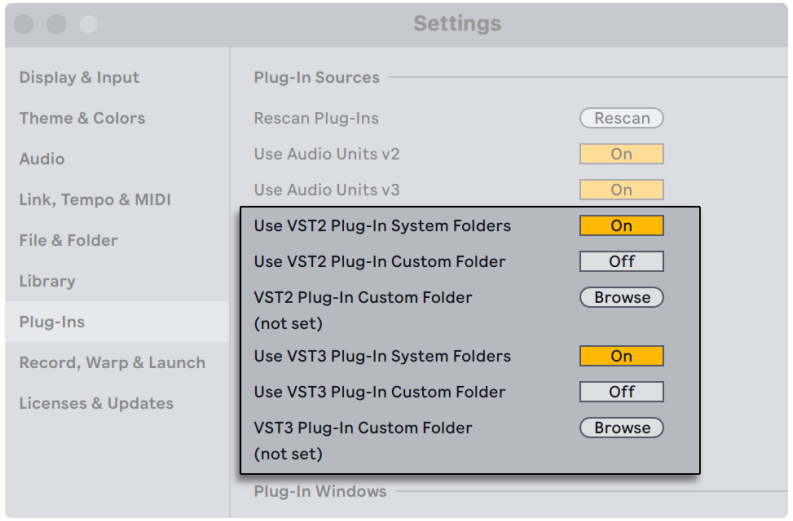
When you start Live for the first time, you will need to activate your VST plug-in sources before working with VST plug-ins. Depending on your computer platform, you may also have to tell Live about the location of the VST plug-in folder containing the devices you want to use. In order to set up your VST sources, press the Activate button in the browser's Plug-Ins panel, or open the Plug-Ins Settings by pressing `Ctrl` `,` (Win) / `Cmd` `,` (Mac). There you will find the Plug-In Sources section.



The VST Plug-In Sources for Windows.

For Windows, proceed as follows:

1. Use the VST Plug-In Custom Folder entry to tell Live about the location of your VST plug-ins: Click the *Browse* button to open a folder-search dialog for locating and selecting the appropriate folder.
2. Once you have selected a VST Custom Folder and Live has scanned it, the path will be displayed. Note that on Windows, Live may have found a path in the registry without the need for browsing.
3. Make sure that the Use VST Plug-In Custom Folder option is set to "On," so that your selected folder is an active source for VST plug-ins in Live. Note that you can choose not to use your VST plug-ins in Live by turning off the Use VST Plug-In Custom Folder option.



The VST Plug-In Sources for macOS.

Set up your VST plug-ins under macOS by doing the following:

1. Your VST plug-ins will normally be installed in the following folder in your home and local directories: /Library/Audio/Plug-Ins/VST. You can turn Live's use of these plug-ins on or off with the Use VST plug-ins in System Folders option.
2. You may have an alternative folder in which you store your VST plug-ins (perhaps those that you use only with Live). You can use VST plug-ins in this folder in addition to, or instead of, those in the System folders. To tell Live about the location of this folder, click the Browse button next to the VST Plug-In Custom Folder entry to open a folder-search dialog for locating and selecting the appropriate folder.
3. Note that you can turn off your VST plug-ins in this folder using the Use VST Plug-In Custom folder option.

Once you have configured your Plug-Ins Settings, the browser will display all plug-ins it finds in the selected VST plug-in folder(s) as well as any sub-folders.

It is also possible to use VST plug-ins stored in different folders on your computer. To do this, create a macOS or Windows *alias* of the folder where additional VST plug-ins are stored, and then place the alias in the VST Plug-In Custom folder (or in the VST Plug-In System folder on macOS) selected in Live's Plug-Ins Settings. The alias can point to a different partition or hard drive on your computer. Live will scan the set VST plug-in folder as well as any alias folders contained therein.

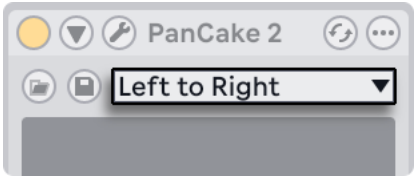
Some VST plug-ins contain errors or are incompatible with Live. During the scanning process, these may cause the program to crash. When you re-launch Live, a dialog will appear to inform you about which plug-in caused the problem. Depending on what Live detects about the plug-in, you may be given the choice between performing another scan or making the problematic plug-in unavailable. If you choose to rescan and they crash the program a second time, Live will automatically make them

unavailable, meaning that they will not appear in the browser and will not be rescanned again until they are reinstalled.

23.4.2 VST Presets and Banks

Every VST plug-in instance "owns" a *bank of presets*. A preset is meant to contain one complete set of values for the plug-in's controls.

To select a preset from the plug-in's bank, use the chooser below the title bar. The number of presets per bank is fixed. You are always working "in" the currently selected preset, that is, all changes to the plug-in's controls become part of the selected preset.



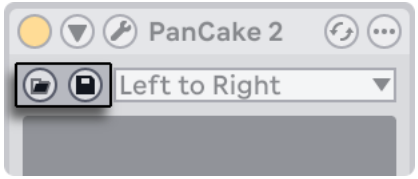
The VST Plug-In Preset Chooser.

Note that VST presets are different from Live device presets: Whereas the presets for a Live device are shared amongst all instances and Live Sets, the VST presets "belong" to this specific instance of the VST plug-in.

To rename the current preset, select the VST's Device Title Bar and execute the Edit menu's Rename Plug-In Preset command. Then type in a new preset name and confirm by pressing `Enter` .

Renaming a VST Plug-In Preset.

VST presets and banks can be imported from files. Clicking a VST's Load Preset or Bank button brings up a standard file-open dialog for locating the desired file.



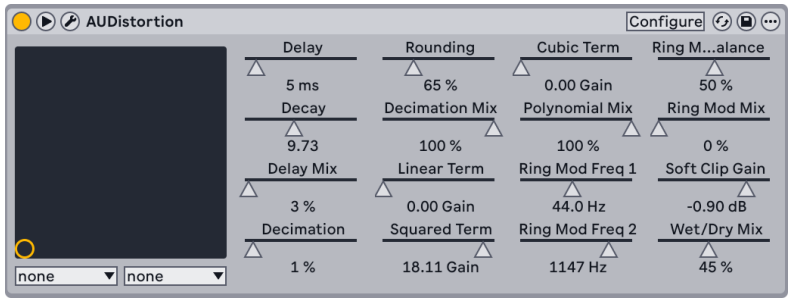
The VST Load Preset or Bank Button (Left) and Save Preset or Bank Button (Right).

Windows only: Please select from the File Type filter in the Windows dialog whether you want to locate VST Presets (VST Program Files) or VST Banks (VST Bank Files).

To save the currently selected preset as a file, click the VST Save Preset or Bank button to bring up a standard file-save dialog; select "VST Preset" from the Format menu (Mac)/from the File Type menu (Windows); select a folder and name. For saving the entire bank as a file, proceed likewise but choose "VST Bank" as a file type/format.

23.5 Audio Units Plug-Ins

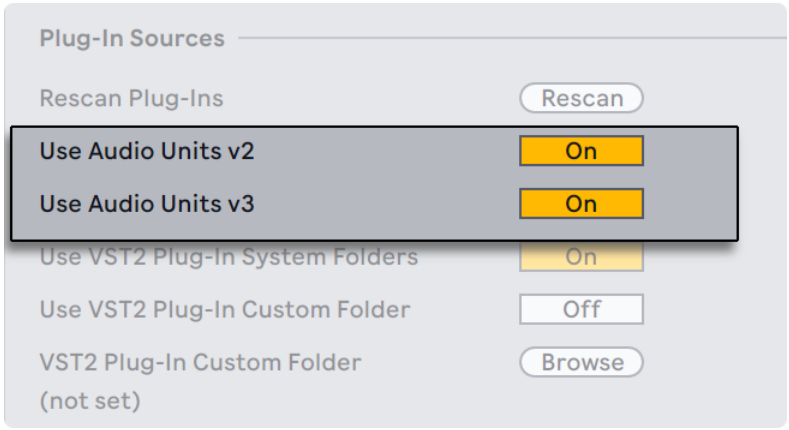
Audio Units plug-ins are only available in macOS. In most respects, they operate just like VST plug-ins.



An Audio Units Plug-In.

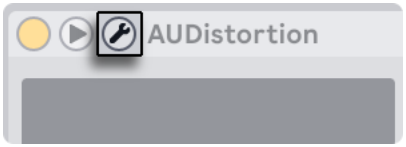
The first time you open Live, Audio Units plug-ins will not appear in the browser. In order to activate your Audio Units as a plug-in source, please press the Activate button in the browser's Plug-Ins label, or go to the Plug-Ins Settings by pressing `Ctrl` , (Win) / `Cmd` , (Mac). There you will find the Plug-In Sources section. Turning on the Use Audio Units option activates Audio Units plug-ins so that they appear in Live's browser.

Note that you can always turn this option off later if you decide not to use Audio Units.



Activating Audio Units Plug-Ins.

Audio Units plug-ins sometimes have a feature that allows choosing between different modes for the device. You might be able to choose, for example, between different levels of quality in the rendering of a reverb. Choosers for these device modes can only be accessed through the original plug-in panel, which is opened using the Show/Hide Plug-In Window button.



Opening an Audio Units Plug-In Window.

Audio Units have presets that function just like those for the Live effects. However, some AU presets cannot be dragged to different locations in the browser, as they are read-only.

Audio Units presets have an .au preset extension and are stored in the following directory according to their manufacturer's name:

(Home)/Library/Audio/Presets/(Manufacturer Name)/(Plug-in Name)

23.6 Device Delay Compensation

Live automatically compensates for delays caused by Live and plug-in instruments and effects, including those on the return tracks. These delays can arise from the time taken by devices to process an input signal and output a result. The compensation algorithm keeps Live's tracks in sync while minimizing delay between the player's actions and the audible result.

Device delay compensation is on by default and does not normally have to be adjusted in any way. To manually turn latency compensation on (or off), use the Delay Compensation option in the Options menu.

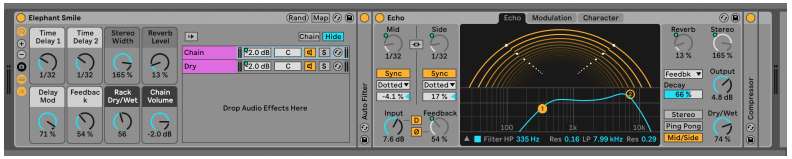
When Delay Compensation is on, the Reduced Latency When Monitoring option is available in the Options menu. This option toggles latency compensation on and off for tracks which have input monitoring on. When enabled, input-monitored tracks will have the lowest possible latency, but may be out of sync with some other tracks in your Set, such as Return tracks, which are still delay compensated. When disabled, all tracks will be in sync, but input-monitored tracks may have higher latency.

Note that tempo-synced effects and other devices that get timing information from Live's internal clock may sound out of sync if they are placed in a device chain after devices which cause delay.

Unusually high individual track delays or reported latencies from plug-ins may cause noticeable sluggishness in the software. If you are having latency-related difficulties while recording and playing back instruments, you may want to try turning off device delay compensation; however, this is not normally recommended. You may also find that adjusting the individual track delays is useful in these cases, but please note that the Track Delay controls are unavailable when device delay compensation is deactivated.

Note that device delay compensation can, depending on the number of tracks and devices in use, increase the CPU load.

24. Instrument, Drum and Effect Racks

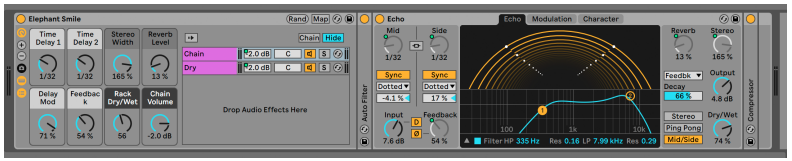


Inside An Audio Effect Rack.

A Rack is a flexible tool for working with effects, plug-ins and instruments in a track's device chain. Racks can be used to build complex signal processors, dynamic performance instruments, stacked synthesizers and more. Yet they also streamline your device chain by bringing together your most essential controls. While Racks excel at handling multiple devices, they can extend the abilities of even a single device by defining new control relationships between its parameters.

24.1 An Overview of Racks

24.1.1 Signal Flow and Parallel Device Chains



Inside An Audio Effect Rack.

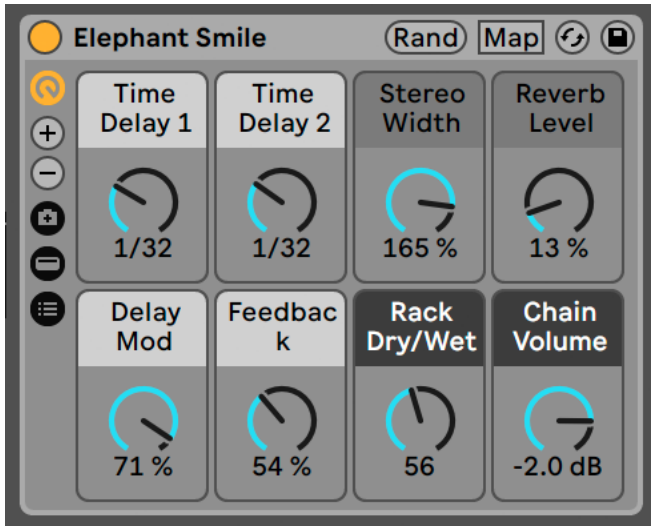
In any of Live's tracks, devices are connected serially in a device chain, passing their signals from one device to the next, left to right. By default, the Device View displays only a single chain, but there is actually no limit to the number of chains contained within a track.

Racks allow (among other things) additional device chains to be added to any track. When a track has multiple chains, they operate in *parallel*: In Instrument and Effect Racks, each chain receives the same input signal at the same time, but then processes its signal *serially* through its own devices. The output of each of the parallel chains is mixed together, producing the Rack's output.

Drum Racks also allow multiple parallel chains to be used simultaneously, but their chains process input somewhat differently: Rather than receiving the same input signals, each Drum Rack chain receives input from only a single assigned MIDI note.

The entire contents of any Rack can be thought of as a single device. This means that adding a new Rack at any point in a device chain is no different than adding any other device, and Racks can contain any number of other Racks. If more devices are placed after a Rack in a track's device chain, the Rack's output is passed on to them, as usual.

24.1.2 Macro Controls



The Macro Controls.

One unique property of Racks are their Macro Controls.

The Macro Controls are a bank of knobs, each capable of addressing any number of parameters from any devices in a Rack. How you use them is up to you — whether it be for convenience, by making an important device parameter more accessible; for defining exotic, multi-parameter morphs of rhythm and timbre; or for constructing a mega-synth, and hiding it away behind a single customized interface. See Using the Macro Controls for a detailed explanation of how to do this.

For the greatest degree of expression, try MIDI-mapping the Macro Controls to an external control surface.

24.2 Creating Racks

Four Rack variants cover the range of Live's devices: Instrument Racks, Drum Racks, Audio Effect Racks and MIDI Effect Racks. Just as with track types, each kind of Rack has rules regarding the devices it contains:

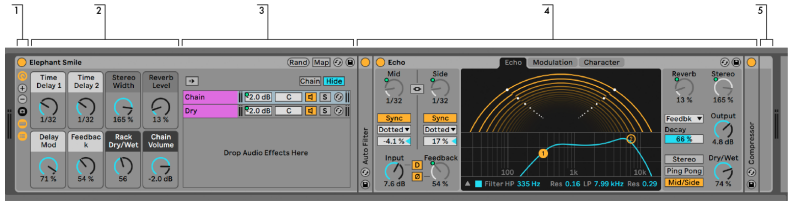
- *MIDI Effect Racks* contain only MIDI effects, and can only be placed in MIDI tracks.
- *Audio Effect Racks* contain only audio effects, and can be placed in audio tracks. They can also be placed in MIDI tracks, as long as they are "downstream" from an instrument.
- *Instrument Racks* contain instruments, but can additionally contain both MIDI and audio effects. In this case, all MIDI effects have to be at the beginning of the Instrument Rack's device chain, followed by an instrument, and then any audio effects.
- *Drum Racks* are similar to Instrument Racks; they can contain instruments as well as MIDI and audio effects and their devices must be ordered according to the same signal flow rules. Drum Racks can also contain up to six return chains of audio effects, with independent send levels for each chain in the main Rack.

There are different ways to create Racks. A new, empty Rack can be created by dragging a generic Rack preset ("*Audio Effect Rack*," for example) from the browser into a track. Devices can then be dropped directly into the Rack's Chain List or Devices view, which are introduced in the next section.

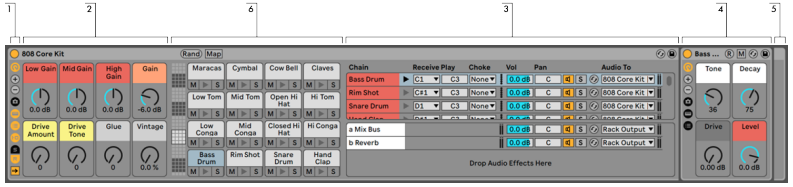
If a track already has one or more devices that you would like to group into a Rack, then simply select the title bars of those devices in the Device View, and right-click on one of the title bars to reveal the *Group* and *Group to Drum Rack* commands in the context menu. Note that if you repeat this command again on the same device, you will create a Rack within a Rack. You can also group multiple chains within a Rack using the same procedure. Doing this also creates a Rack within a Rack. In the Device View, the contents of Racks are always contained between end brackets: Just as with punctuation or in mathematics, a Rack within a Rack will have a pair of brackets within a pair of brackets.

To ungroup devices, dismantling their Racks, select the *Rack's* title bar, and then use the Edit menu or the context menu to access the *Ungroup* command.

24.3 Looking at Racks



Components of an Effect Rack.



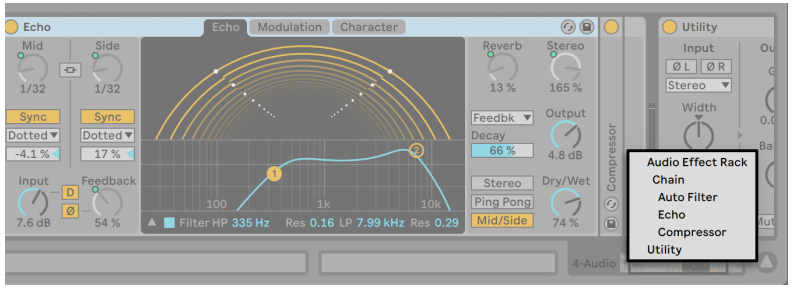
Components of a Drum Rack.

1. Racks have distinct views that can be shown or hidden as needed. Therefore, every Rack has a *view column* on its far left side that holds the corresponding view selectors. The actual view selectors available differ depending on whether an Instrument, Drum or Effect Rack is being used.
2. *Macro Controls*
3. *Chain List* - In Drum Racks, this view can include both drum chains and return chains.
4. *Devices*
5. Racks are also identifiable by their round corners, which bracket and enclose their content. When the Devices view is shown, the *end bracket* visually detaches itself to keep the Rack hierarchy clear.
6. *Pad View* - This is unique to Drum Racks.

To move, copy or delete an entire Rack at once, simply select it by its title bar (as opposed to the title bars of any devices that it contains). When selected, a Rack can also be renamed by using the Edit menu's *Rename* command. You can also enter your own info text for a Rack via the Edit Info Text command in the Edit menu or in the Racks's context menu.

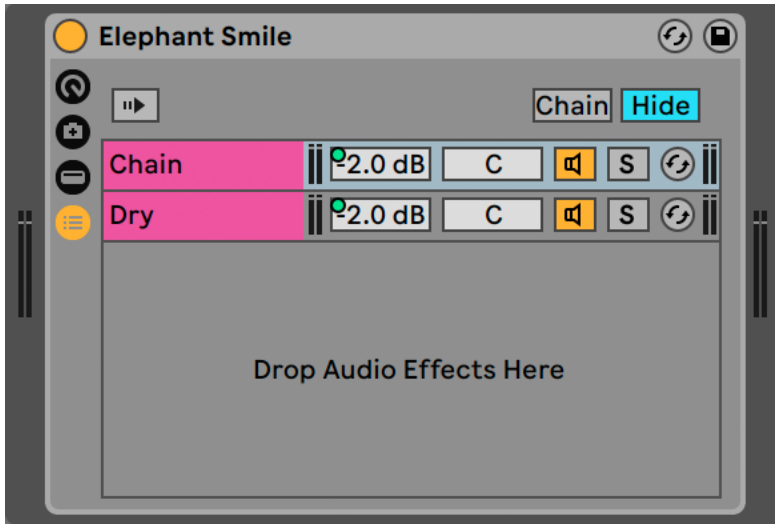
When all of a Rack's views are hidden, its title bar will fold into the view column, making the entire Rack as slim as possible. This has the same effect as choosing Fold from the context menu or double-clicking on the Rack's title bar.

If you would like to locate a particular device in a Rack without searching manually through its entire contents, you will appreciate this navigation shortcut: right-click on the Device View selector, and a hierarchical list of all devices in the track's device chain will appear. Simply select an entry from the list, and Live will select that device and move it into view for you.



Navigate Racks Quickly Via a Context Menu.

24.4 Chain List



The Chain List in an Audio Effect Rack.

As signals enter a Rack, they are first greeted by the Chain List. We will therefore also choose this point for our own introduction.

The Chain List represents the branching point for incoming signals: Each parallel device chain starts here, as an entry in the list. Below the list is a drop area, where new chains can be added by dragging and dropping presets, devices, or even pre-existing chains.

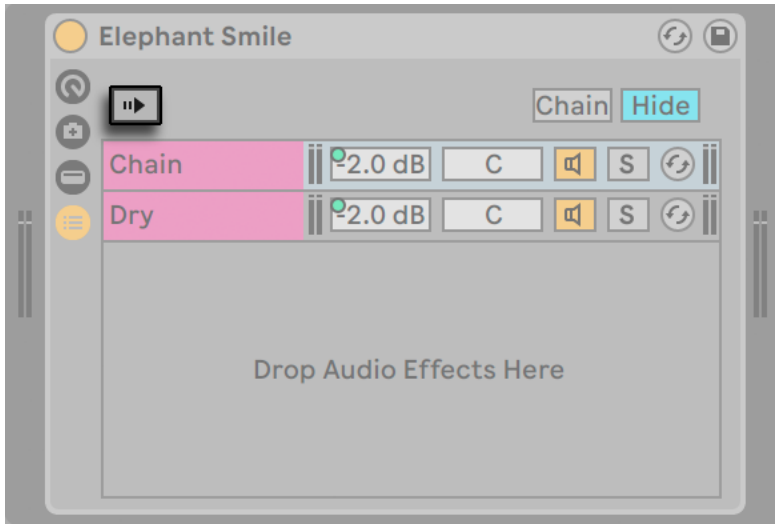
Note: Racks, chains and devices can be freely dragged into and out of other Racks, and even between tracks. Selecting a chain, then dragging and hovering over another Session or Arrangement View track will give that track focus; its Device View will open, allowing you to drop your chain into place.

Since the Device View can show only one device chain at a time, the Chain List also serves as a navigational aid: The list selection determines what will be shown in the adjacent Devices view (when enabled). Try using your computer keyboard's up and down arrow keys to change the selection in the Chain List, and you'll find that you can quickly flip through the contents of a Rack.

The Chain List also supports multi-selection of chains, for convenient copying, organizing and regrouping. In this case, the Devices view will indicate how many chains are currently selected.

Each chain has its own Chain Activator, as well as Solo and Hot-Swap buttons. Chains in Instrument, Drum and Audio Effect Racks also have their own volume and pan sliders, and Drum Rack chains have additional send level and MIDI assignment controls. Like Live Clips, entire chains can be saved and recalled as presets in the browser. You can give a chain a descriptive name by selecting it, then choosing the Edit menu's Rename command. You can also enter your own info text for a chain via the Edit Info Text command in the Edit menu or in the chain's context menu. The context menu also contains a color palette where you can choose a custom chain color.

24.4.1 Auto Select



Auto Select in an Audio Effect Rack.

When the Auto Select switch is activated, every chain that is currently processing signals becomes selected in the Chain List. In Drum Racks, this feature will select a chain if it receives its assigned MIDI input note. In Instrument and Effect Racks, Auto Select works in conjunction with zones, which are discussed next, and is quite helpful when troubleshooting complex configurations.

24.5 Zones

Zones are sets of data filters that reside at the input of every chain in an Instrument or Effect Rack. Together, they determine the range of values that can pass through to the device chain. By default, zones behave transparently, never requiring your attention. They can be reconfigured, however, to form sophisticated control setups. The three types of zones, whose editors are toggled with the buttons above the Chain List, are *Key*, *Velocity*, and *Chain Select*. The adjacent Hide button whisks them out of sight.

Note: Audio Effect Racks do not have key or velocity zones, since these two zone types filter MIDI data only. Likewise, Drum Racks have no zones at all; they filter MIDI notes based on choosers in their chain lists.

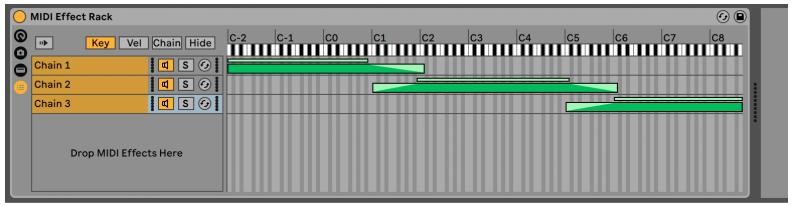
Zones contain a lower, main section, used for resizing and moving the zone itself, and a narrow upper section that defines fade ranges. Resizing of either section is done by clicking and dragging on its right or left edges, while moving is accomplished by clicking and dragging a zone from anywhere except its edges.

24.5.1 Signal Flow through Zones

To understand how zones work, let's examine the signal flow in a MIDI Effect Rack. Our MIDI Effect Rack resides in the device chain of a MIDI track, and therefore processes MIDI signals. We will assume that it contains four parallel device chains, each containing one MIDI effect.

1. All MIDI data in the track is passed to its device chain, and therefore into the input of the MIDI Effect Rack.
2. Our MIDI Effect Rack has four device chains, all of which receive the same MIDI data at the same time.
3. Before any MIDI data can enter a device chain, it must be able to pass through every zone in that chain. Every chain in a MIDI Effect Rack has three zones: a key zone, a velocity zone and a chain select zone.
4. An incoming MIDI note gets compared to a chain's key zone. If the MIDI note lies within the key zone, it is passed to the next zone for comparison; if it does not, then we already know that the note will not be passed to that chain's devices.
5. The same comparisons are made for the chain's velocity and chain select zones. If a note also lies within both of these zones, then it is passed to the input of the first device in that chain.
6. The output of all parallel chains is mixed together to produce the MIDI Effect Rack's final output. If there happened to be another device following after the Rack in the track's device chain, it would now receive the Rack's output for processing.

24.5.2 Key Zones

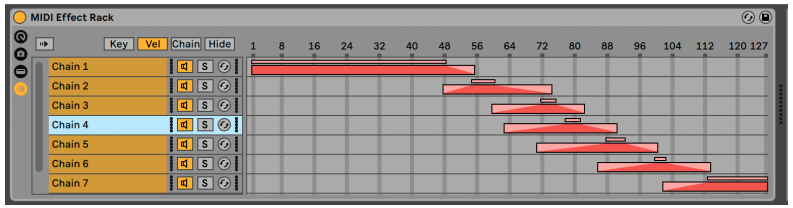


The Key Zone Editor.

When the Key button is selected, the Key Zone Editor appears to the right of the Chain List, illustrating how each chain maps to the full MIDI note range (nearly 11 octaves). Chains will only respond to MIDI notes that lie within their key zone. The zones of individual chains may occupy any number of keys, allowing for flexible "keyboard split" setups.

Key zone fade ranges attenuate the *velocities* of notes entering a chain.

24.5.3 Velocity Zones



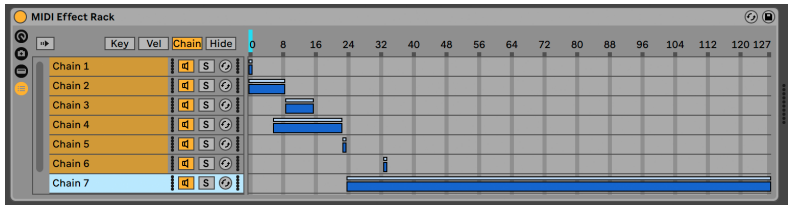
The Velocity Zone Editor.

Each chain in an Instrument Rack or MIDI Effect Rack also has a velocity zone, specifying the range of MIDI Note On velocities that it will respond to.

The Velocity Zone Editor, when displayed, replaces the Key Zone Editor to the right of the Chain List. MIDI Note On velocity is measured on a scale of 1-127, and this value range spans the top of the editor. Otherwise, the functionality here is identical to that of the Key Zone Editor.

Velocity zone fade ranges attenuate the velocities of notes entering a chain.

24.5.4 Chain Select Zones



The Chain Select Editor.

Activating the Chain button in an Instrument or Effect Rack displays the Chain Select Editor. These Racks have chain select zones, which allow you to filter chains spontaneously via a single parameter. The editor has a scale of 0-127, similar to the Velocity Zone Editor. Above the value scale, however, you will find a draggable indicator known as the Chain selector.

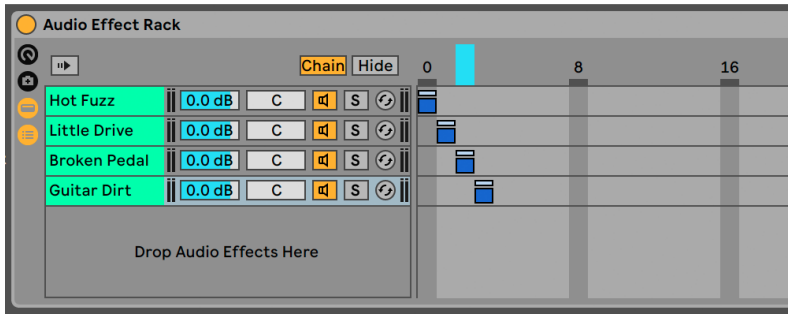
The chain select zone is a data filter just like the other zones; although all chains in a Rack receive input signals, only those with chain select zones that overlap the current value of the Chain selector can be addressed and thereby produce output.

By default, the chain select zones of Instrument and MIDI Effect Racks filter only notes, ignoring all other incoming MIDI events (such as MIDI CCs). To filter all MIDI events, enable the Chain Selector Filters MIDI Ctrl option, available in the context menu of a Rack's Chain Select Ruler.

In MIDI Effect Racks, fade ranges attenuate the velocities of notes entering a chain. In Instrument Racks and Audio Effect Racks, which both output audio signals, fade ranges attenuate the volume level at each chain's output. So what happens, then, if the Chain selector is moved outside of the chain select zone where a sound is currently playing? If the zone ends in a fade range, the chain's output volume is attenuated to zero while the Chain selector is outside of the zone. If the zone had no fade range, the output volume is *not* attenuated, allowing the chain's effects (like long reverb tails or delays) to fade out according to their own settings.

Let's consider how we can make use of chain select zones in a performance situation:

24.5.4.1 Making Preset Banks Using Chain Select



Using Chain Select Zones to Create Effects Presets.

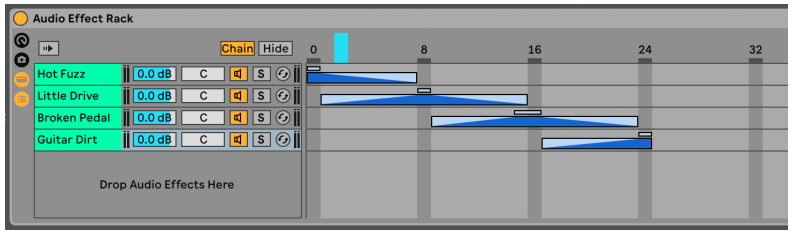
Unlike the other zone types, the default length of a chain select zone is 1, and the default value is 0. From this setup, we can quickly create "preset banks" using the Chain Select Editor.

Again, we will use a Rack with four chains as our starting point. Each of the four chains contain different effects that we would like to be able to switch between. To make this a "hands-on" experience, we have MIDI-mapped the Chain selector to an encoder on an external control surface.

Let's move the chain select zones of the second and third chains so that each of our zones is occupying its own adjacent value: The first chain's zone has a value of 0, the second chain's zone has a value of 1, the third has a value of 2, and the fourth has a value of 3.

Since each of our chain select zones has a unique value, with no two zones overlapping, we now have a situation where only one chain at a time can ever be equal to the Chain selector value (shown at the top of the editor). Therefore, by moving the Chain selector, we determine which chain can process signals. With our MIDI encoder at hand, we can now flip effortlessly between instrument or effect setups.

24.5.4.2 Crossfading Preset Banks Using Fade Ranges



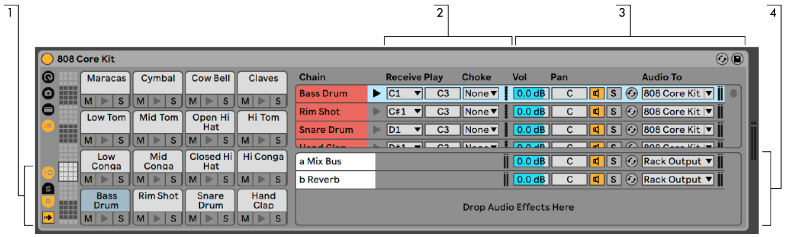
Crossfading Between Effects Presets Using Chain Select Zones.

Taking the previous example one step further, we can tweak our chain select zones to produce a smooth transition between our "presets." To accomplish this, we will make use of our zones' *fade ranges*.

To create some room for fading, let's extend the length of our zones a bit. Setting the zones as shown maintains four exclusive values for our presets, so that each still has one point where neither of the others are heard. We crossfade between the presets over eight steps. If this is too rough a transition for your material, simply reposition the zones to maximize the fade ranges.

24.6 Drum Racks

We've already talked a bit about Drum Racks, and most of their features are the same as those found in Instrument and Effect Racks. But Drum Racks have a slightly different layout, some unique controls and special behavior that is optimized for creating drum kits.



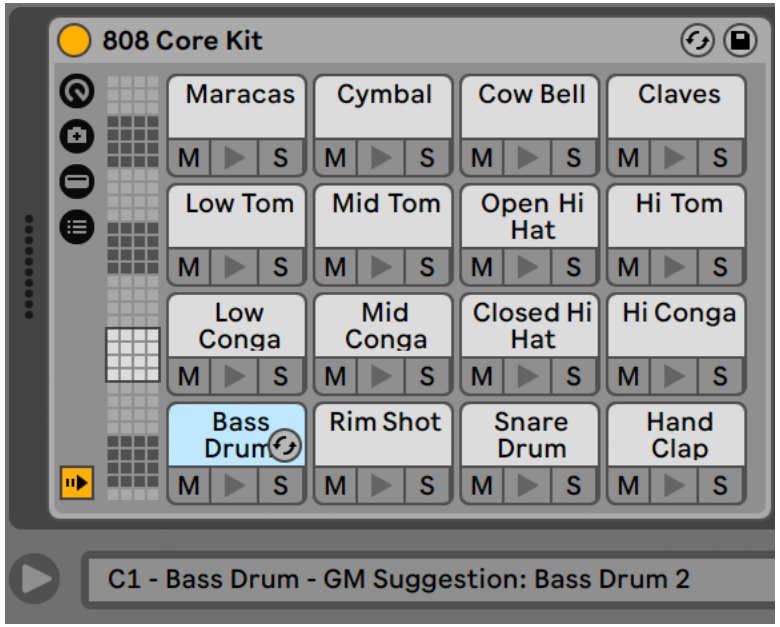
The Chain List in a Drum Rack.

1. In addition to the standard selectors found on all Racks, Drum Racks have four additional controls in the view column. From top to bottom, these are toggles for the Input/Output, Send, and Return sections, and the Auto Select button.
2. *Input/Output Section* - The Receive chooser sets the incoming MIDI note to which the drum chain will respond. The list shows note names, MIDI note numbers and standard GM drum equivalents. The Play slider sets the outgoing MIDI note that will be sent to the devices in the chain. The Choke chooser allows you to set a chain to one of sixteen choke groups. Any chains that are in the same choke group will silence the others when triggered. This is useful for choking open hihats by triggering closed ones, for example. If "All Notes" is selected in the Receive chooser, the Play and Choke choosers are disabled — in this case, the chain simply passes the note that it receives to its devices. The small Preview button to the left of these choosers fires a note into the chain, making it easy to check your mappings away from a MIDI controller.
3. *Mixer Section* - In addition to the mixer and Hot-Swap controls found in other Rack types, Drum Racks also have send sliders. These sliders allow you to set the amount of post-fader signal sent from each drum chain to any of the available return chains. Note that send controls are not available until return chains have been created.

4. *Return Chains* - A Drum Rack's return chains appear in a separate section at the bottom of the chain list. Up to six chains of audio effects can be added here, which are fed by send sliders in each of the drum chains above.

The Audio To chooser in the mixer for return chains allows you to route a return chain's output to either the main output of the Rack or directly to the return tracks of the Set.

24.6.1 Pad View



Pad View.

The Pad View is unique to Drum Racks and offers an easy way to map and manipulate samples and devices. Each pad represents one of the 128 available MIDI notes. The pad overview to the left shifts the set of visible pads up or down in groups of 16, either by dragging the view selector to a new area or by using your computer keyboard's up and down arrow keys. Use the `Alt` (Win) / `Cmd` (Mac) modifier to shift the view by single rows instead.

Almost any object from Live's browser — samples, effects, instruments and presets — can be dragged onto a pad, mapping automatically to the pad's note and creating or reconfiguring internal chains and devices as necessary. Dropping a sample onto an empty pad, for example, creates a new chain containing a Simpler, with the dropped sample ready to play from the pad's note. If you then drag an audio effect to the same pad, it is placed downstream from the Simpler in the same chain. To replace the Simpler, simply drop another sample onto the same pad — any downstream audio effects or upstream MIDI effects will be left intact and only the Simpler and sample will be replaced.

In addition to dragging objects from the browser, pads can also be filled quickly via Hot-Swap. If you're in Hot-Swap mode, pressing the `D` key will toggle the Hot-Swap target between the Drum Rack itself and the last selected pad.

If a multi-selection of samples is dropped onto a pad, new Simplers and chains will be mapped upwards chromatically from this pad, replacing any other samples that may have already been assigned to the pads in question (but, as before, leaving any effects devices alone). `Alt` (Win) / `Cmd` (Mac)-dragging a multi-selection layers all of the samples to a single pad, by creating a nested Instrument Rack.

Dragging a pad to another pad swaps the note mapping between the pads. This means that any MIDI clips triggering the affected notes will now play the "wrong" sounds — although this might be exactly what you want. `Alt` (Win) / `Cmd` (Mac)-dragging one pad to another will layer any chains from both pads in a nested Instrument Rack.

You can always change your mappings from within the chain list as well, by adjusting the Receive choosers. The Pad View will update automatically to reflect your changes. If you set the same Receive note for multiple chains, that note's pad will trigger them all.

If you're working with lots of nested Racks, the inner structure can quickly become complicated. Pad View can make it much easier to work by letting you focus on only the top level: the notes and sounds. It's important to remember that a pad represents a note, rather than a chain. More specifically, it represents all chains, no matter how deep in the Rack, that are able to receive that pad's note. What you can control with each pad is related to how many chains it represents:

- An empty pad shows only the note it will trigger. When you mouse over it, the Status Bar will display this note, as well as the suggested GM instrument.
- A pad that triggers only one chain shows the name of the chain. In this case, the pad serves as a handy front-end for many controls that are normally accessed in the chain list, such as mute, solo, preview and Hot-Swap. You can also rename and delete the chain via the pad.
- A pad that triggers multiple chains shows "Multi" as its name, and its mute, solo and preview buttons will affect all of its chains. If you mute and solo chains individually within the chain list, the pad's icons reflect this mixed state. Hot-Swap and renaming are disabled for a Multi pad, but you can delete all of its chains at once.

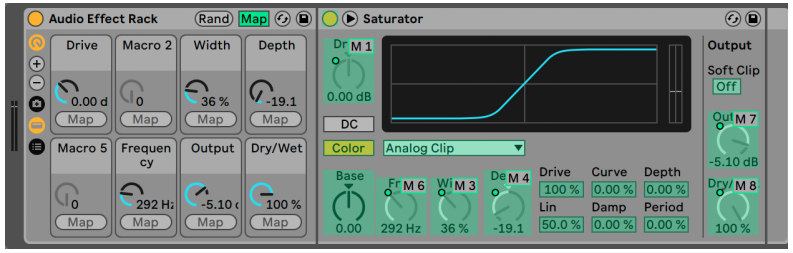
Although Pad View is designed for easy editing and sound design, it also excels as a performance interface, particularly when triggered by a hardware control surface with pads. If your pad controller is one of Live's natively supported control surfaces, simply select it as a control surface in the Link, Tempo & MIDI tab of Live's Settings. From then on, as long as you have a Drum Rack on a track that's receiving MIDI, your pad controller will trigger the pads that are visible on your screen. If you scroll the pad overview to show a different set of pads, your controller will update automatically.

24.7 Using the Macro Controls

It is possible to use up to 16 Macro Controls in a Rack. When creating a new Rack, eight Macro Control knobs are shown by default. You can use the ⊕ and ⊖ view selector buttons to increase or decrease the number of visible Macro Controls. Note that the state of shown and hidden Macro Controls is saved in the Live Set.

These Selector Buttons Set the Number of Visible Macro Controls.

24.7.1 Map Mode



Making Macro Control Assignments in Map Mode.

With the potential for developing complex device chains, Macro Controls keep things manageable by taking over the most essential parameters of a Rack (as determined by you, of course). Once you have set up your ideal mapping, the rest of the Rack can be hidden away.

The Macro Control view's dedicated Map button opens the door to this behavior. Enabling Macro Map Mode causes three things to happen:

- All mappable parameters from a Rack's devices will appear with a colored overlay;
- Map buttons will appear beneath each Macro Control dial;
- The Mapping Browser (see 'The Mapping Browser') will open.

The following steps will get you started mapping:

1. Enable Macro Map Mode by clicking the Map mode button;
2. Select a device parameter for mapping by clicking it once;
3. Map the parameter by clicking on any Macro Control's Map button. The details will be added to the Mapping Browser. By default, the Macro Control will take its name and units from the device parameter it is controlling.
4. Refine the value range if desired using the Min/Max sliders in the Mapping Browser. Inverted mappings can be created by setting the Min slider's value greater than the Max slider's value. The current values can also be inverted by right-clicking on the entry in the Mapping Browser and selecting Invert Range.
5. Select another device parameter if you'd like to create more mappings, or click on the Map mode button once more to exit Macro Map Mode.

Note that once assigned to a Macro Control, a device parameter will appear disabled, since it hands over all control to the Macro Control (although it can still be modulated externally, via clip envelopes.

You can edit or delete your assignments at any time using the Mapping Browser (which only appears when Map Mode is enabled).

If more than one parameter is assigned to a single Macro Control, the name of the control will revert to its generic name (e.g., Macro 3). The Marco Control's units will also change to a 0 to 127 scale, except when all parameters possess both the same unit type and the same unit range.

Macro Controls can be given custom names, colors and info text entries via the corresponding commands in the Edit menu or the context menu.

24.7.2 Randomizing Macro Controls

If you want to add an element of surprise or find some inspiration in your Set, randomizing Macro Controls can be a useful tool. You can randomize the values of all mapped Macro Controls in a Rack by pressing the Rand button in that Rack's title bar.

This Button Randomizes Macro Controls.

Depending on your material, you might only want to randomize some parameters, while leaving other controls unchanged. To exclude a mapped Macro Control from randomization, enable the Exclude Macro from Randomization option in the context menu. Note that Macro Controls assigned to Volume parameters in Instrument Rack presets are excluded from randomization by default.

24.7.3 Macro Control Variations

Macro Control Variations.

You can store different states of Macro Controls as individual presets (or "variations"). This is useful when, for example, you want to capture the state of a Rack as a "snapshot" during a sound design session, or audition different settings of a mapped Macro Control in an Audio Effect Rack made for mixing. You can also use these variations or create builds and drops, or make instant jumps between different Macro Control settings while recording or performing.

Clicking the Show/Hide Macro Variations view selector button opens a view where you can store, manage and launch Macro Control variations.

This Selector Button Opens the Macro Control Variations View.

Pressing the New button stores the current state of a Rack's Macro Controls as a new variation. By default, each stored variation will be named sequentially as "Variation 1", "Variation 2", etc. Selected Macro Control variations can be renamed, duplicated, or deleted via commands in the Edit menu or the context menu.

A Macro Control variation can be launched in its stored state via the Launch Macro Variation button to the right, or overwritten via the "Overwrite Macro Variation" button to the left. Note that you can exclude a Macro Control from changing when a different Macro Control variation is launched, by using the Exclude Macro From Variations command in the context menu. Disabling the context menu entry will re-enable changes to that control.

24.8 Mixing With Racks

Any Instrument or Drum Rack that contains more than one chain can be viewed and mixed alongside the tracks in the Session View's mixer. A track that contains these Racks will have a ⏷ button in its title bar, which will fold the Rack's mixer in or out. Likewise, any nested chains within the Rack will also have this button. This makes it easy to get an overview of your Rack's hierarchy — or hide it when you just want to work on your mix.

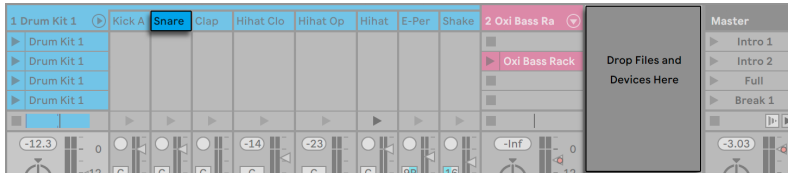
Mixing Rack Chains in the Session View.

Chains in the Session View mixer look similar to tracks, but they have no clip slots. Their mixing and routing controls mirror those found in the Rack's chain list, so any changes made to these controls in either area will be reflected in the other immediately. Likewise, many chain operations such as reordering, renaming and regrouping can be performed from either the mixer or the chain list. Clicking in a chain's mixer title bar shows only that chain's devices in the Device View.

As with tracks, when chains are multiselected in the Session View mixer, adjusting a mixer parameter for one of the chains will adjust the same parameter in the other selected chains. Note that this behavior only applies to parameters adjusted via the Session mixer and not when adjusting the same parameters in the Rack's chain list.

24.8.1 Extracting Chains

All chains can be dragged from their parent Racks and placed into other tracks or Racks, either from the chain list or from the Session View mixer. A Drum Rack's return chains can also be extracted, and will create new return tracks if dragged to the mixer. Drum chains have an additional feature: when dragged from the mixer to a new track, they take their MIDI notes with them. For example, if you are working on a MIDI drum loop within a single track and decide that you would like to move just the snare onto its own track, simply select the snare chain's title bar in the mixer and drag it to the mixer's drop area. This creates a new track with the full contents of the snare chain: both its devices and its MIDI data. If you would like to extract only the devices, drag from the chain list instead of from the mixer.



Extracting Drum Chains in the Mixer Extracts MIDI Data.